

DATA MINING PROJECT REPORT

Predicting Employee Burnout in the Tech Industry: A Multi-Model Machine Learning Approach

Course: Data Mining (BA360)

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Abbreviation Table

EDA	Exploratory Data Analysis
KNN	K-Nearest Neighbors
LR	Logistic Regression
PCA	Principal Component Analysis
PC1	Principal Component 1
HR	Human Resources

Executive Summary

Employee burnout has become one of the most pressing challenges facing technology companies in the modern workplace. Prolonged exposure to high workloads, elevated stress levels, and insufficient organizational support has been linked to significant reductions in productivity, increased absenteeism, and high staff turnover; all of which carry measurable financial and operational consequences.

This project addresses burnout prediction as a data mining classification problem, applying a comprehensive machine learning pipeline to a dataset of approximately 150,000 tech-sector employee records.

Five supervised classification algorithms were evaluated: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and Naive Bayes. All models achieved test accuracy in the range of 89–93%, with Logistic Regression and Random Forest performing best. Feature importance analysis identified stress level, depression score, weekly working hours, anxiety score, and work-life balance as the five most influential predictors of burnout.

Exploratory analysis through correlation heatmaps, box plots, and pairplots confirmed strong relationships between psychological indicators and burnout outcomes. Dimensionality reduction via Principal Component Analysis (PCA) revealed that high-burnout employees are meaningfully separable in two-dimensional feature space, suggesting the existence of coherent underlying behavioral and psychological profiles.

The findings of this study carry direct implications for human resources management and organizational strategy. By deploying predictive models in real time; using routinely collected HR and wellness survey data; organizations can proactively identify at-risk employees, deploy targeted interventions, and ultimately reduce the human and financial costs associated with burnout.

1. Drive Link:

https://drive.google.com/drive/folders/1c_8wKKcV7ZmxKLfuOkcPyKASo4h8MqHr?usp=sharing

2. Demo Link:

https://drive.google.com/file/d/19DMOuTu6j1-u60wB6uSBQhMwjB8-0UnV/view?usp=share_link

3. Presentation Link:

<https://canva.link/odo32wc2j6ohkyi>

1. Introduction

Burnout, defined by the World Health Organization as a syndrome resulting from chronic workplace stress that has not been successfully managed, represents a growing public health and organizational concern.

In the technology sector, where demanding project cycles, long working hours, and high-performance expectations are commonplace, burnout rates have consistently exceeded those observed in other industries.

The consequences are well-documented: impaired cognitive function, disengagement, increased healthcare utilization, and accelerated departure from employment.

Against this backdrop, data-driven approaches to early burnout detection have gained considerable traction. Machine learning models, when trained on behavioral, psychological, and organizational data, can surface patterns that are invisible to traditional management reporting and HR analytics. The emergence of large-scale employee wellness datasets; often collected through workplace surveys, performance management systems, and occupational health programs; has made it feasible to build predictive systems capable of flagging burnout risk before it manifests in critical behavioral outcomes.

This project was motivated by a desire to apply data mining techniques to a real-world, socially relevant problem. Rather than focusing on a single algorithmic approach, the study benchmarks five distinct classifiers across a common pipeline, enabling a rigorous comparative evaluation. The project further incorporates exploratory data analysis, feature engineering, dimensionality reduction, and cluster-based employee segmentation to provide a holistic analytical perspective.

The remainder of this report is structured as follows: Section 2 describes the dataset; Section 3 formalizes the problem statement; Section 4 details data preparation procedures; Section 5 describes the analytical methods; Section 6 presents and interprets results; Section 7 discusses limitations and improvement pathways; and Section 8 draws concluding remarks.

2. Objectives

This project was designed to fulfill the following primary and secondary objectives:

2.1 Primary Objectives

- To build and evaluate machine learning classification models capable of predicting employee burnout level (low, medium, high) using workplace and psychological variables.
- To identify the key drivers of burnout through feature importance analysis and correlation examination.
- To provide actionable, business-oriented insights that can inform HR strategy and employee wellness programs.

2.2 Secondary Objectives

- To compare the predictive performance of five distinct classification algorithms: Logistic Regression, Decision Tree, Random Forest, KNN, and Naive Bayes.
- To apply dimensionality reduction (PCA) and unsupervised clustering (K-Means) to enrich the analytical understanding of employee burnout profiles.
- To assess the distributional characteristics and inter-variable relationships present in the dataset through comprehensive exploratory data analysis.
- To illustrate the importance of avoiding data leakage and applying sound preprocessing practices in any applied machine learning pipeline.

3. Main research questions:

RQ1: Which individual, psychological, and organizational factors are the most significant predictors of employee burnout in the technology sector?

This question drives the feature importance analysis. Rather than assuming all variables contribute equally, the study seeks to identify which specific factors; among stress level, depression score, work hours, manager support, work-life balance, and others ; carry the strongest predictive weight. Answering this question helps organizations understand *where* burnout originates, not just *who* is affected.

RQ2: Can machine learning classification models accurately predict an employee's burnout level (low, medium, high) based on observable behavioral and psychological indicators?

This is the core predictive question of the project. It evaluates whether supervised learning algorithms (Logistic Regression, Decision Tree, Random Forest, KNN, and Naive Bayes) can reliably classify employees into three burnout categories using

non-leaking features. The answer determines whether a deployable early-warning system is technically feasible.

RQ3: Which classification algorithm offers the best predictive performance for burnout detection, and does model complexity provide a measurable advantage over simpler approaches?

This comparative question benchmarks five algorithms against each other. A key sub-question is whether ensemble methods like Random Forest justify their computational complexity over interpretable linear models like Logistic Regression. The answer has practical deployment implications, particularly in HR environments where model transparency matters.

RQ4: Do employees at different burnout levels exhibit distinct, separable profiles in the high-dimensional feature space, and can dimensionality reduction techniques reveal these patterns visually?

This question is addressed through PCA. If high- and low-burnout employees occupy distinct regions even in two-dimensional reduced space, it confirms that the feature set encodes genuine discriminative structure; validating the modeling approach and supporting the existence of meaningful burnout archetypes rather than a diffuse, unpredictable phenomenon.

RQ5: To what extent do relational and organizational support factors; such as managerial support, social support, and work-life balance; moderate burnout risk, and can their influence be quantified?

This question shifts focus from *stressors* to *protective factors*. While psychological burden drives burnout, the study examines whether strong managerial support or healthy work-life balance can buffer against it. Quantifying the moderating effect of these variables provides actionable levers that organizations can adjust through policy and management practice.

RQ6: Is employee burnout risk uniformly distributed across demographic groups (age, gender, job role), or is it primarily determined by working conditions and psychological state?

This equity-relevant question examines whether burnout is a structural problem tied to working conditions or a demographic one. The near-zero correlation of demographic variables with burnout level in the dataset suggests the latter, but this must be explicitly tested and stated. The answer shapes how organizations should target their wellness initiatives; universally or at specific groups.

3. Methodology

3.1 Dataset Description

The dataset used in this study was sourced from Kaggle and is titled the "Tech Mental Health & Burnout Dataset." It comprises approximately 150,000 observations, each representing a unique tech-sector employee. The data was collected through a simulated but realistic survey framework designed to reflect the kinds of information organizations might obtain through employee wellness questionnaires and HR records.

Dataset_Link:

<https://www.kaggle.com/datasets/suhanigupta04/employee-mental-health-and-burnout-dataset>

Each record contains 25 variables spanning four conceptually distinct domains:

Demographic Variables

- age: Employee age in years (range: approximately 22–55).
- gender: Categorical variable (encoded numerically).
- job_role: Role classification within the tech organization (e.g., developer, analyst, manager).
- experience_years: Total years of professional experience.

Organizational Variables

- company_size: Categorical indicator of the employing organization's scale.
- work_mode: Remote, hybrid, or on-site classification.
- work_hours_per_week: Average weekly hours worked.
- overtime_hours: Additional hours worked beyond contracted hours.
- meetings_per_day: Average number of meetings attended daily.
- deadlines_missed: Count of missed project deadlines.
- job_satisfaction: Self-reported satisfaction with current role (scale: 1–10).
- manager_support: Perceived quality of managerial support (scale: 1–10).
- work_life_balance: Self-assessed balance between work and personal life (scale: 1–10).

Behavioral and Lifestyle Variables

- sleep_hours: Average daily sleep duration.
- physical_activity_days: Days per week of physical exercise.
- screen_time_hours: Daily non-work screen time.
- caffeine_intake: Daily caffeine consumption units.
- social_support_score: Perceived social support outside of work.
- has_therapy: Binary indicator of current engagement in therapeutic support.
- seeks_professional_help: Binary indicator of proactive help-seeking behavior.

Psychological and Target Variables

- `stress_level`: Self-reported stress level (scale: 1–10).
- `anxiety_score`: Anxiety assessment score.
- `depression_score`: Depression assessment score.
- `burnout_score`: A continuous composite burnout measure (1–8 scale). Excluded from modeling to prevent data leakage.
- `burnout_level`: The ordinal target variable (0 = low, 1 = medium, 2 = high).

3.2 Data Preparation

3.2.1 Missing Value Treatment

Missing values were handled differently depending on the variable type. For numerical columns (integers and floats), missing entries were replaced with the median of each column; a robust measure that is not distorted by outliers. For categorical columns (object type), missing values were filled with the mode, i.e. the most frequently occurring category. This two-step approach ensures that each variable type is treated in the most statistically appropriate way while keeping the dataset complete for modeling.

```
# Fill missing values
```

```
num_cols = df.select_dtypes(include=['int64', 'float64']).columns
df[num_cols] = df[num_cols].fillna(df[num_cols].median())
cat_cols = df.select_dtypes(include=['object']).columns
df[cat_cols] = df[cat_cols].fillna(df[cat_cols].mode().iloc[0])
```

3.2.2 Categorical Encoding

All categorical variables—including `gender`, `job_role`, `company_size`, and `work_mode`—were transformed using Label Encoding, which assigns a unique integer to each category level. While this approach is straightforward, it is worth noting that for tree-based algorithms (Decision Tree, Random Forest), label encoding does not distort model learning because splits are made on thresholds. For distance-based models (KNN, Logistic Regression), this encoding was acceptable given the subsequent application of feature scaling.

```
# Encode categorical columns
```

```
le = LabelEncoder()
for col in cat_cols:
    df[col] = le.fit_transform(df[col])
```

3.2.3 Feature Scaling

For algorithms sensitive to the magnitude of input features—specifically Logistic Regression, KNN, and Naive Bayes—features were standardized using `StandardScaler`, which transforms each feature to have zero mean and unit variance. Tree-based models (Decision Tree and Random Forest) do not require scaling, as they operate through recursive binary splits rather than distance or gradient computations.

```
# Scale for models that need scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

3.2.4 Feature Selection and Leakage Prevention

A critical preprocessing decision involved the removal of `burnout_score` from the feature matrix. This variable is a direct composite proxy for `burnout_level`—indeed, the correlation heatmap confirms a coefficient of approximately 0.71 between the two. Including it in model training would constitute data leakage: the model would effectively be learning to predict the target variable from a near-equivalent representation of itself, producing artificially inflated accuracy figures that would not generalize to real-world deployment where only behavioral and survey inputs are available.

```
# Features and target
X = df.drop(['burnout_level', 'burnout_score'], axis=1)
y = df['burnout_level']
```

The final feature set comprised 23 variables, excluding `burnout_score` and the target `burnout_level`.

Data was partitioned into an 80% training set and a 20% test set using stratified splitting with a fixed random seed of 42 to ensure reproducibility.

```
# Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

3.3 Analytical Methods

3.3.1 Exploratory Data Analysis (EDA)

Prior to modeling, a comprehensive EDA was conducted to understand variable distributions, inter-feature relationships, and class composition. Key EDA components included: a histogram of the `burnout_score` distribution to assess its skewness; a box plot comparing `burnout_score` distributions across `burnout_level` categories to validate the target variable's discriminative power; a full correlation heatmap to identify multicollinearity and variable-target associations; and a pairplot focusing on the key quantitative variables (`burnout_score`, `age`, `work_hours_per_week`, `stress_level`).

```
df['burnout_score'].hist(bins=20)
plt.title("Distribution of Burnout Score")
plt.xlabel("Burnout Score")
plt.ylabel("Frequency")
plt.show()
```

3.3.2 Classification Models

To address the core objective of predicting **burnout_level (0 = low, 1 = medium, 2 = high)**, five supervised learning algorithms were implemented and evaluated on the prepared dataset.

The dataset was split into **80% training and 20% testing sets**, ensuring reproducibility with a fixed random seed. Models were trained either on:

- **Standardized data (`X_train_scaled`)** for distance- and gradient-based models, or
- **Raw data (`X_train`)** for tree-based models, consistent with their theoretical properties.

Logistic Regression: Logistic Regression was implemented as a **baseline linear classifier** to evaluate whether burnout is linearly separable in the feature space.

- Input: standardized features
- Parameters: `max_iter = 1000` (to ensure convergence due to dataset size)
- Output: class probabilities and predicted burnout category

This model is particularly relevant in this project because:

- It provides **interpretable coefficients**, allowing identification of key drivers (e.g., `stress_level`, `work_hours_per_week`),
- It aligns with the hypothesis suggested by EDA and PCA that **burnout classes are approximately linearly separable**.

Logistic Regression

```
model_lr = LogisticRegression(max_iter=1000)
model_lr.fit(X_train_scaled, y_train)
y_pred_lr = model_lr.predict(X_test_scaled)
print("\nLogistic Regression")
print("Accuracy:", accuracy_score(y_test, y_pred_lr))
print(classification_report(y_test, y_pred_lr))
```

Decision Tree: A Decision Tree classifier was trained to capture **non-linear relationships and interaction effects** between variables such as:

- stress_level × work_hours_per_week
- manager_support × work_life_balance
- Input: non-scaled features
- Parameter: `max_depth = 5` to control overfitting and maintain interpretability

The depth constraint was intentionally imposed to:

- Avoid memorization of the training data,
- Produce a model that can be interpreted by HR stakeholders through **decision rules**.

Decision Tree

```
model_dt = DecisionTreeClassifier(max_depth=5, random_state=42)
model_dt.fit(X_train, y_train)
y_pred_dt = model_dt.predict(X_test)
print("\nDecision Tree")
print("Accuracy:", accuracy_score(y_test, y_pred_dt))
print(classification_report(y_test, y_pred_dt, zero_division=0))
```

Random Forest: Random Forest was implemented as the primary **ensemble learning method**, combining multiple decision trees to improve predictive performance.

- Input: non-scaled features
- Parameters: `n_estimators = 100, random_state = 42`

In this project, Random Forest serves two key purposes:

1. **Improving prediction accuracy** by reducing variance compared to a single tree,
2. Providing **feature importance scores**, which are critical for answering Research Question 1 (identifying burnout drivers).

As shown later (Figure 5, page 18), this model confirms that **stress_level, depression_score, and work_hours_per_week** dominate prediction.

```
# Random Forest
```

```
model_rf = RandomForestClassifier(n_estimators=100, random_state=42)
```

```
model_rf.fit(X_train, y_train)
```

```
y_pred_rf = model_rf.predict(X_test)
```

```
print("\nRandom Forest")
```

```
print("Accuracy:", accuracy_score(y_test, y_pred_rf))
```

```
print(classification_report(y_test, y_pred_rf, zero_division=0))
```

K-Nearest Neighbors (KNN): KNN was implemented as a **distance-based, non-parametric classifier** to evaluate how well burnout categories can be inferred from similarity between employees.

- Input: standardized features
- Parameter: $k = 5$

This model relies on the assumption that:

Employees with similar behavioral and psychological profiles (e.g., stress, sleep, work hours) are likely to share similar burnout levels.

However, due to the large dataset size, KNN is:

- Computationally expensive at prediction time,
- Highly sensitive to feature scaling, which justifies the use of StandardScaler.

```
# KNN
```

```
model_knn = KNeighborsClassifier(n_neighbors=5)
```

```
model_knn.fit(X_train_scaled, y_train)
```

```
y_pred_knn = model_knn.predict(X_test_scaled)
```

```
print("\nKNN")
```

```
print("Accuracy:", accuracy_score(y_test, y_pred_knn))
```

```
print(classification_report(y_test, y_pred_knn, zero_division=0))
```

Naive Bayes (Gaussian): Gaussian Naive Bayes was applied as a **probabilistic baseline model**, assuming that features follow a normal distribution and are conditionally independent given the burnout class.

- Input: standardized features

Despite its simplifying assumptions, this model is valuable in this project because:

- It provides **fast probabilistic predictions**,
- It performs reasonably well in high-dimensional datasets like ours (23 features).

```
# Naive Bayes
```

```
model_nb = GaussianNB()
```

```
model_nb.fit(X_train_scaled, y_train)
```

```
y_pred_nb = model_nb.predict(X_test_scaled)
```

```
print("\nNaive Bayes")
```

```
print("Accuracy:", accuracy_score(y_test, y_pred_nb))
```

```
print(classification_report(y_test, y_pred_nb, zero_division=0))
```

Model Evaluation Strategy

All models were evaluated on the test set using:

- **Accuracy** (overall performance),
- **Classification report (precision, recall, F1-score)**,
- **Confusion matrices** to analyze misclassification patterns across burnout levels.

Special attention was given to:

- **Recall of the high-burnout class**, as it represents the most critical group for intervention,
- The ability of models to handle **class imbalance**, identified during EDA.

3.3.3 Dimensionality Reduction: PCA

Principal Component Analysis (PCA) was applied to the **standardized feature space** to reduce dimensionality and evaluate the intrinsic structure of the dataset.

- Input: scaled full feature set (excluding target and leakage variable)
- Components: `n_components = 2`

Unlike a purely theoretical use of PCA, in this project it serves a **diagnostic and validation role**:

1. **Visualization of burnout class separability**

The projection of data into the first two principal components allows us to verify whether employees with different burnout levels form distinguishable clusters.

2. **Validation of modeling feasibility**

As observed in Figure 6 (page 19), burnout classes show **clear separation along the first principal component**, indicating that the dataset contains strong discriminative information.

3. **Support for linear modeling**

The observed structure explains why **Logistic Regression performs nearly as well as Random Forest**, confirming that the decision boundary is largely linear.

Thus, PCA was not used for feature reduction in modeling, but rather as a **visual and analytical tool to validate the structure of the data and support interpretation of model results**.

```
from sklearn.decomposition import PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_train_scaled)

# PCA for visualization
scaler = StandardScaler()
pca = PCA(n_components=2)
X_pca_all = pca.fit_transform(scaler.fit_transform(X))
```

3.3.4 Unsupervised Clustering: K-Means

To complement supervised classification, K-Means clustering was applied to identify **latent employee profiles independent of burnout labels**.

- Input: full feature matrix (after preprocessing)
- Parameter: **n_clusters = 3** (aligned with burnout levels for interpretability)

The clustering results were added to the dataset as a new variable (**cluster**) and used to:

- Explore **behavioral and psychological groupings**,
- Identify **employee archetypes** such as:
 - High-stress, overworked employees
 - Balanced, low-risk employees
 - Moderate-risk transitional profiles

This step is particularly relevant for practical applications, as it enables:

- **Segmentation-based HR strategies,**
- Targeted interventions beyond simple classification.

Unlike classification, clustering does not rely on burnout labels, making it valuable for:

- Detecting hidden structures,
- Validating whether burnout emerges naturally from feature patterns.

```
# fit KMeans and assign labels to df
kmeans = KMeans(n_clusters=3, random_state=42)
df['cluster'] = kmeans.fit_predict(X) # ← this line was missing
```

4. Results

This section presents the outputs of the exploratory and predictive analyses, accompanied by the visualizations generated throughout the project. Each figure is interpreted in the context of its business and analytical significance.

4.1 Exploratory Data Analysis

Figure 1: Distribution of Burnout Score

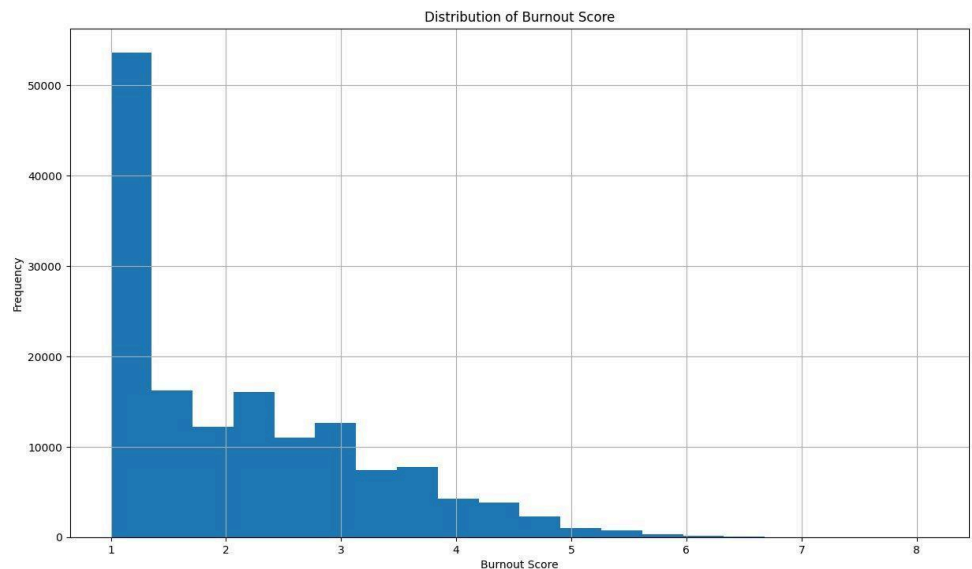


Figure 1: Distribution of Burnout Score

The burnout score distribution is strongly right-skewed, with the majority of employees concentrated at scores between 1 and 2. This indicates that while severe burnout (scores above 4) is relatively rare in absolute terms, a meaningful tail of high-scoring individuals exists.

This skewness has practical implications: any predictive model must be assessed not merely on overall accuracy; which can be inflated by the dominant low-burnout class; but also on its sensitivity to correctly identifying high-burnout cases, which represent the population most in need of organizational intervention.

Figure 3: Correlation Heatmap

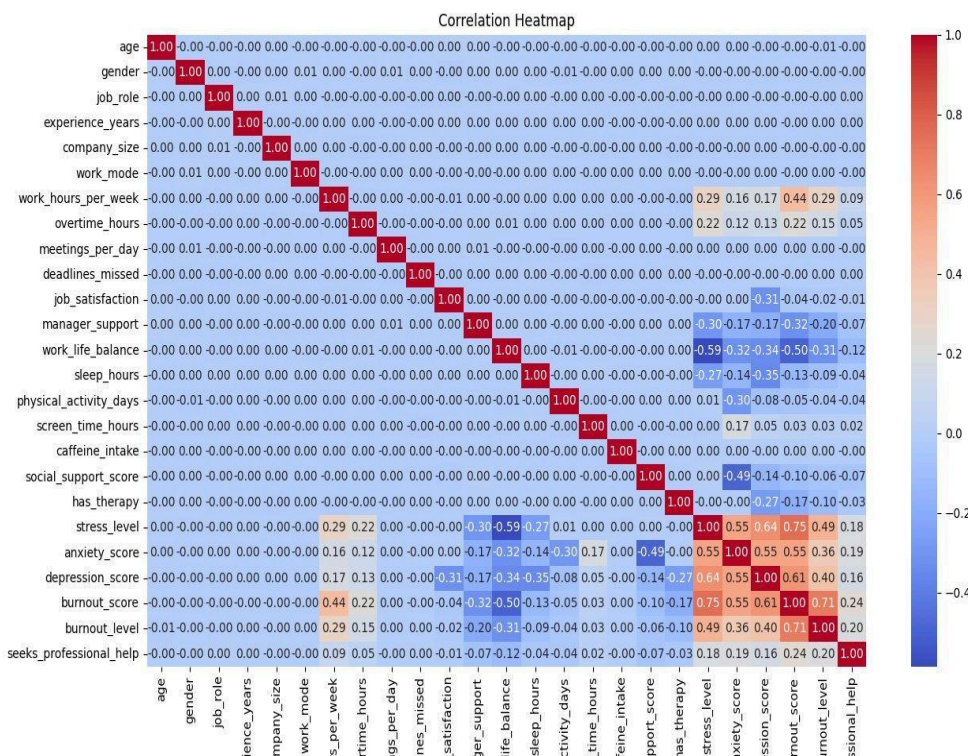
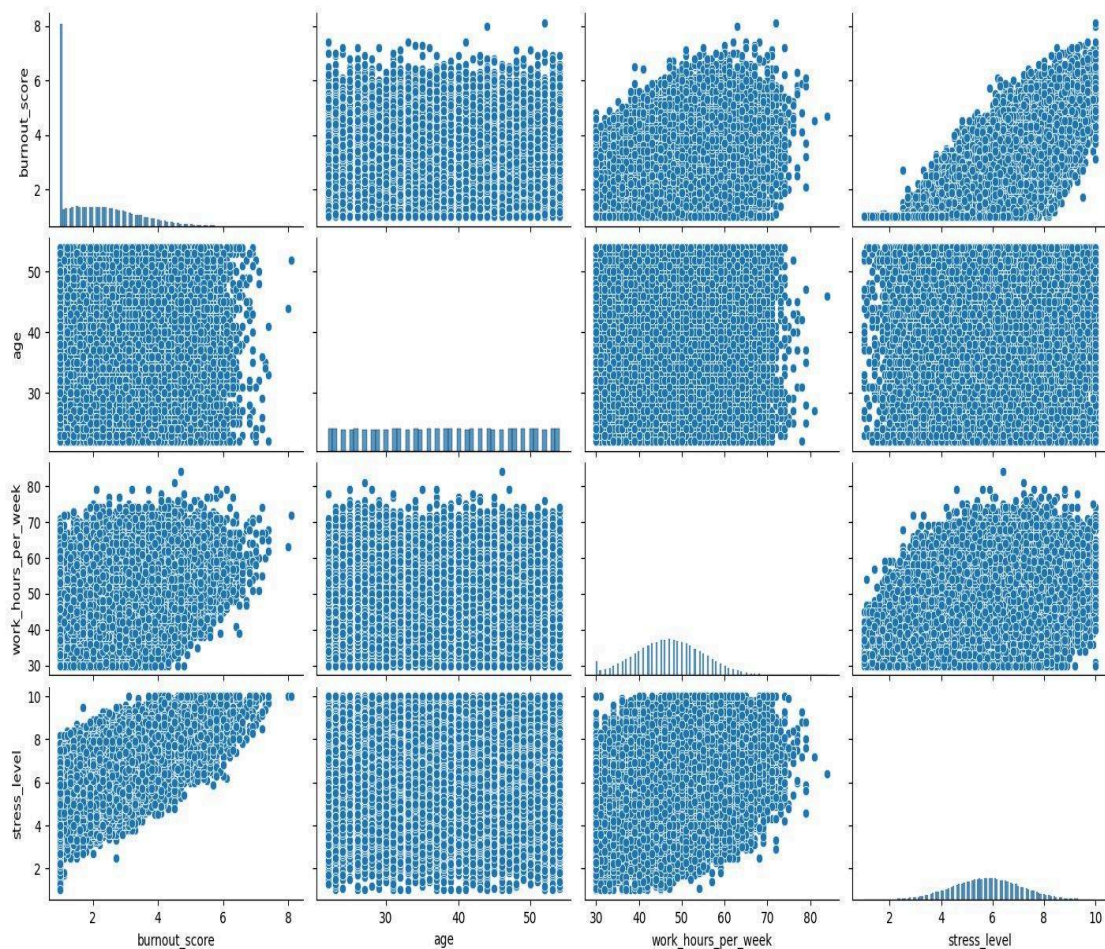


Figure 3: Correlation Heatmap of All Variables

The correlation heatmap reveals several substantively important relationships. The strongest positive correlation with burnout_level is observed for stress_level ($r \approx 0.49$), followed by anxiety_score ($r \approx 0.36$), depression_score ($r \approx 0.40$), and work_hours_per_week ($r \approx 0.29$). Conversely, work_life_balance exhibits a strong negative correlation with burnout_level ($r \approx -0.31$), as does manager_support ($r \approx -0.20$). The correlation between burnout_score and burnout_level ($r \approx 0.71$) further justifies its exclusion from model features. Notably, demographic variables such as age, gender, and job_role show near-zero correlations, suggesting that burnout risk in this dataset is driven more by environmental and psychological factors than by individual characteristics.

Figure 4: Pairplot – Key Quantitative Variables*Figure 4: Pairplot – Burnout Score, Age, Work Hours, and Stress Level*

The pairplot provides a multivariate view of the four most analytically relevant continuous variables. The most striking relationship is the strong positive linear association between `stress_level` and `burnout_score`, which is consistent with the correlation heatmap findings and reinforces the primacy of psychological stress as a burnout precursor. `work_hours_per_week` also shows a positive, though less linear, association with `burnout_score`—employees working in excess of 60 hours per week appear disproportionately represented at higher burnout scores. Age shows little to no pattern with burnout, confirming that burnout is not an age-specific phenomenon in this dataset but rather a function of working conditions and psychological state.

4.2 Classification Results

Figure 5: Random Forest Feature Importance

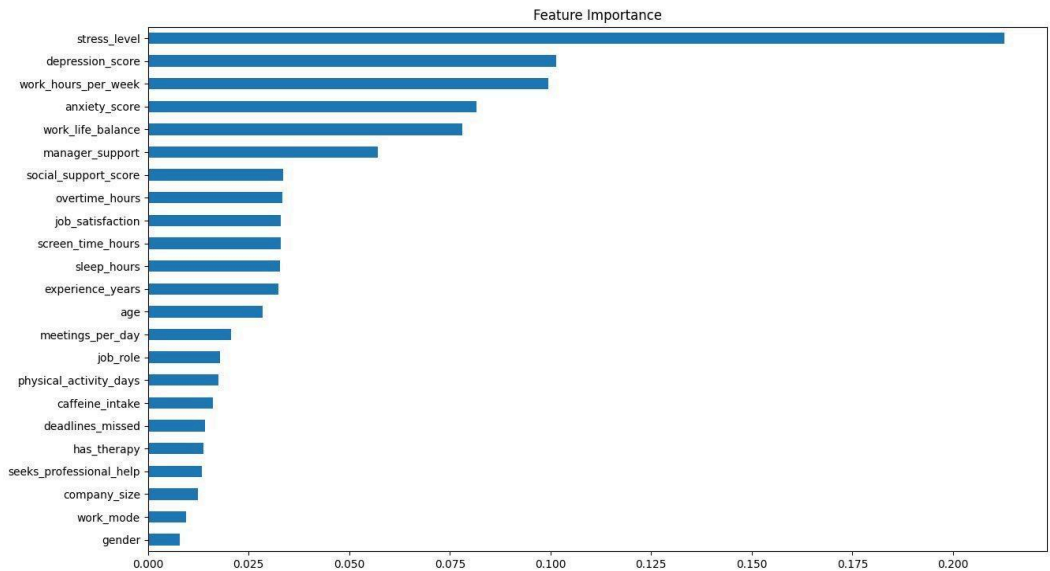


Figure 5: Random Forest Feature Importance

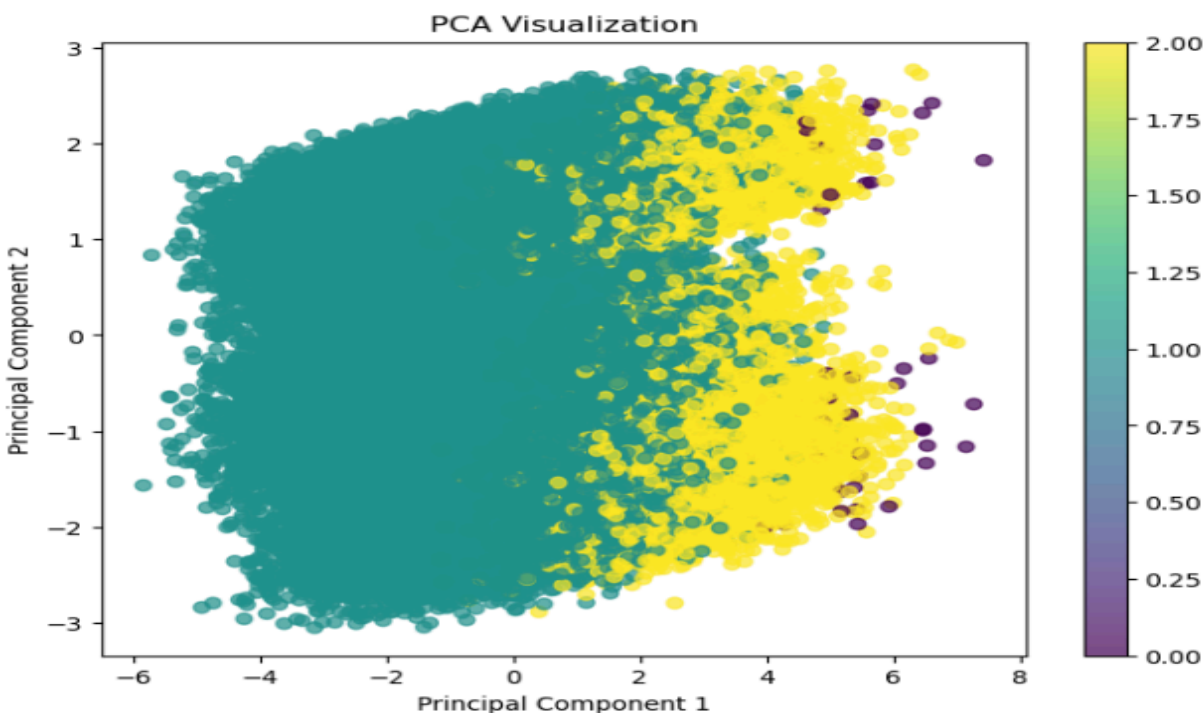
The Random Forest feature importance analysis identifies `stress_level` as by far the most influential predictor, with an importance score approaching 0.21; roughly twice that of the next highest features.

`depression_score` and `work_hours_per_week` follow with approximately equal importance (around 0.10 each), succeeded by `anxiety_score` and `work_life_balance` (approximately 0.08 each).

These five variables collectively account for a substantial portion of the model's predictive signal.

Further down the importance ranking, `manager_support`, `social_support_score`, `overtime_hours`, and `job_satisfaction` contribute meaningfully.

Demographics (`gender`, `work_mode`, `company_size`) appear at the very bottom, reinforcing the finding that contextual and psychological factors dominate burnout prediction.

Figure 6: PCA Visualization*Figure 6: PCA Visualization of Training Data (Colored by Burnout Level)*

The PCA projection of the standardized training data into two dimensions reveals a clear separation between burnout classes along PC1.

Low-burnout employees (teal/dark green) are concentrated in the left portion of the plot (negative PC1 values), while high-burnout employees (yellow) are distributed to the right (positive PC1 values). Medium-burnout instances occupy the transitional center zone, with some overlap between groups. A small cluster of dark-purple points (likely representing a rare or extreme class) is visible in the far-right periphery. This spatial structure confirms that the feature set encodes genuine discriminative information and that linear separability, while imperfect, is meaningful—lending credibility to the performance of Logistic Regression alongside ensemble methods.

Confusion Matrices

The following confusion matrices display the prediction distributions for each of the three main models evaluated. In all cases, the target labels are: 0 = low burnout (rare in the test set), 1 = medium burnout (dominant class), and 2 = high burnout (moderate representation).

Figure 7: Random Forest Confusion Matrix

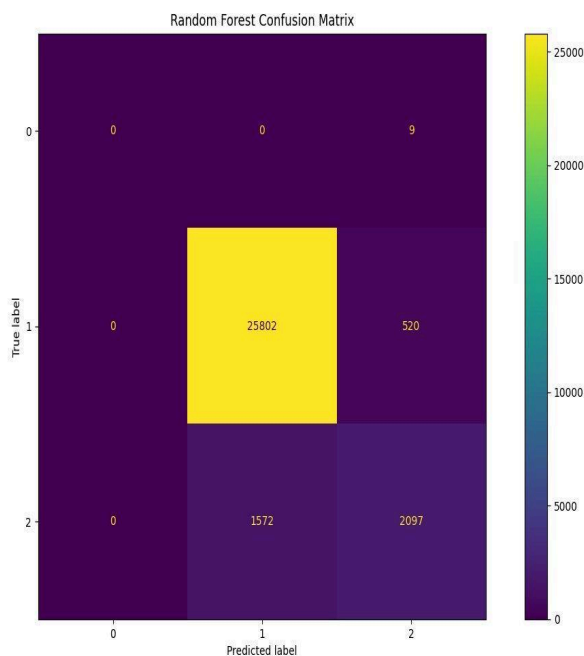


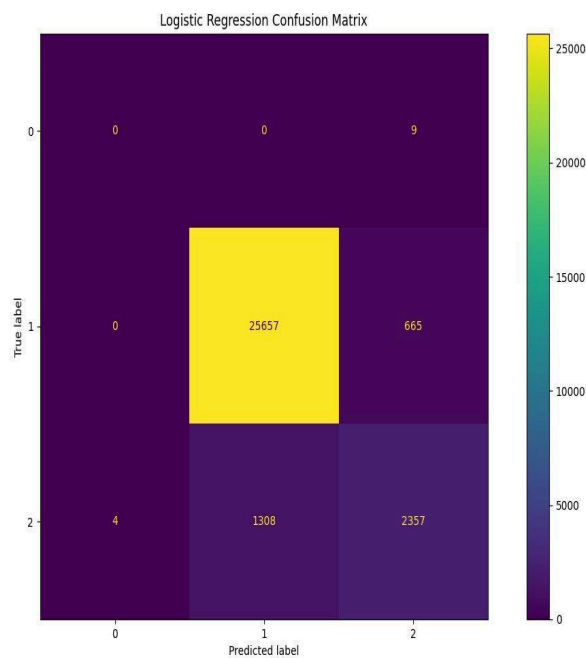
Figure 7: Random Forest Confusion Matrix

The Random Forest model correctly classified the vast majority of medium-burnout instances (25,802 correct), with 520 misclassified as high burnout.

For high-burnout employees, 2,097 were correctly identified versus 1,572 misclassified as medium burnout; a recall challenge for the high-risk class that warrants attention.

All 9 low-burnout instances in the test set were predicted as high burnout, reflecting the near-absence of this class.

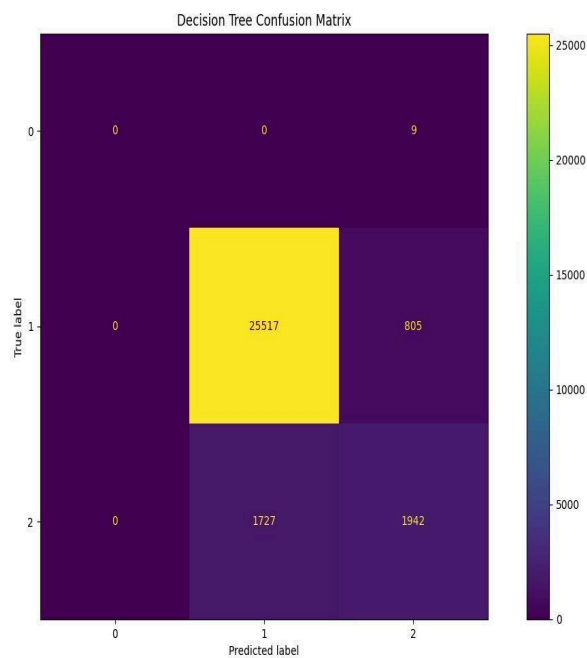
Overall, the model demonstrates strong performance on the dominant class and competitive performance on the high-burnout segment.

Figure 8: Logistic Regression Confusion Matrix*Figure 8: Logistic Regression Confusion Matrix*

Logistic Regression achieved broadly comparable results: 25,657 medium-burnout instances were correctly classified, with 665 misclassified as high.

For the high-burnout class, 2,357 were correctly identified versus 1,308 misclassified as medium; suggesting that LR actually achieves slightly higher recall for the high-burnout class relative to Random Forest, despite its simpler linear structure.

The 4 low-burnout instances categorized as low burnout represent a minor improvement. This result is noteworthy because it suggests that in this particular feature space, linear decision boundaries are nearly as effective as ensemble methods.

Figure 9: Decision Tree Confusion Matrix*Figure 9: Decision Tree Confusion Matrix*

The Decision Tree shows slightly reduced performance: 25,517 medium-burnout predictions were correct, with 805 misclassified as high burnout.

The high-burnout class saw 1,942 correct predictions versus 1,727 misclassified as medium; a lower recall than either Logistic Regression or Random Forest.

The model's depth constraint (`max_depth=5`) limits its capacity to capture complex interaction effects, explaining its slightly lower overall accuracy compared to the ensemble approach.

Figure 10: Model Accuracy Comparison

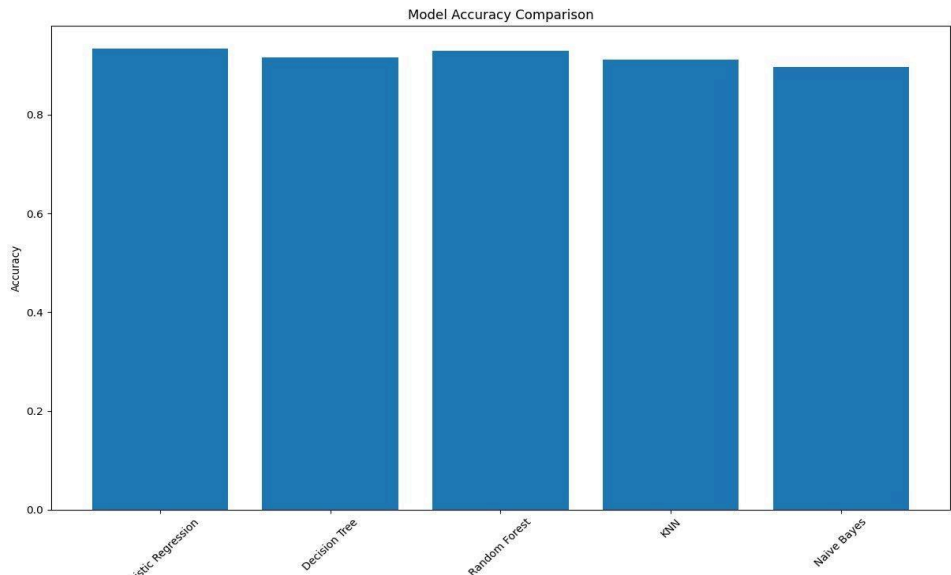


Figure 10: Model Accuracy Comparison Across All Five Classifiers

The accuracy comparison chart summarizes the overall test performance of all five models. Logistic Regression achieved the highest accuracy at approximately 93%, closely followed by Random Forest at approximately 93% as well. Decision Tree and KNN achieved around 91% and 91% respectively, while Naive Bayes recorded the lowest accuracy at approximately 89%. The narrow spread between all models (89–93%) is notable: it indicates that the dataset contains strong, relatively linear signal, making it amenable to simpler models as well as complex ones. However, from an operational standpoint, the additional interpretability provided by Random Forest feature importances makes it the preferred deployment candidate, particularly for HR-facing applications where model transparency is valued.

4.3 Summary of Model Performance

Model	Accuracy	Notes
Logistic Regression	~93%	Highest accuracy; strong linear separability
Random Forest	~93%	Best feature interpretability; robust ensemble
Decision Tree	~91%	Constrained depth; lower recall on class 2
KNN	~91%	Distance-based; computationally expensive
Naive Bayes	~89%	Probabilistic baseline; lowest performer

5. Insights

5.1 Psychological Burden Is the Primary Driver of Burnout

The single most important finding of this analysis is that psychological indicators; stress level, depression score, and anxiety score; collectively explain a far greater proportion of burnout variance than any organizational or demographic variable. Stress level alone accounts for approximately 21% of the Random Forest's total feature importance. This has a direct implication for organizational strategy: investments in mental health support, psychological safety, and stress management programs are likely to yield higher returns in burnout reduction than purely structural interventions such as reorganizations or role redesigns.

5.2 Work Hours and Work-Life Balance Are Critical Organizational Levers

Beyond psychological variables, `work_hours_per_week` and `work_life_balance` are the most organizationally actionable predictors in the model. Employees working over 60 hours per week are disproportionately present in the high-burnout category, and the strong negative correlation between `work_life_balance` and `burnout_level` confirms that balance (rather than mere workload reduction) is the operative factor. Organizations should consider monitoring working-hour distributions systematically and implementing flexible work arrangements tied to measurable balance outcomes.

5.3 Manager Support Matters More Than Peers or Colleagues

Among relational support variables, `manager_support` shows a meaningfully stronger association with burnout outcomes than `social_support_score` (which reflects broader social networks). The feature importance chart ranks `manager_support` sixth overall, ahead of social support, job satisfaction, and lifestyle variables. This suggests that the quality of the direct management relationship is a particularly protective factor; a finding consistent with occupational psychology literature. HR programs targeting frontline manager training in recognition, psychological safety facilitation, and workload mediation are directly supported by this evidence.

5.4 Demographic Variables Are Near-Irrelevant for Burnout Prediction

Gender, age, and `job_role` rank among the lowest-importance features in the Random Forest model and show near-zero correlations with `burnout_level` in the heatmap. This is an important insight for HR equity: it implies that burnout risk is not concentrated in any particular demographic group, but is determined by working conditions and psychological state; factors that organizations can and should actively manage. It also suggests that targeted demographic-based wellness programs may be less cost-effective than universal structural and psychological interventions.

5.5 PCA and Clustering Support the Existence of Coherent Burnout Profiles

The PCA visualization demonstrates that burnout classes occupy distinguishable regions of the feature space, even after compression to two dimensions. This geometric separability suggests that employees at different burnout levels have meaningfully different multivariate profiles—not merely different scores on a single variable. When combined with K-Means clustering, this supports the concept of employee burnout archetypes: groups of employees sharing similar constellations of risk factors. HR analytics teams could operationalize these archetypes to design differentiated intervention strategies rather than one-size-fits-all wellness programs.

5.6 Linear Models Are Competitive with Ensemble Methods in This Domain

The near-parity in accuracy between Logistic Regression ($\approx 93\%$) and Random Forest ($\approx 93\%$) is a practically important finding. It suggests that the decision boundary between burnout classes, in the feature space of this dataset, is approximately linear. This has implications for deployment: a Logistic Regression model is more explainable, faster to retrain, and more easily auditable by HR compliance teams than a Random Forest ensemble. In production environments where transparency is required (for example, when justifying intervention decisions to employees or regulators) Logistic Regression may be preferable despite offering no accuracy advantage.

6. Recommendations

6.1 Deploy a Real-Time Burnout Risk Scoring System

Organizations should consider deploying the trained Logistic Regression or Random Forest model as an ongoing risk scoring engine integrated with HR information systems. By feeding routinely collected survey and HR metrics into the model, managers could receive periodic risk scores for their direct reports—enabling proactive, targeted interventions before burnout manifests in absenteeism or resignation. To ensure ethical deployment, access to individual risk scores should be limited to HR professionals bound by confidentiality obligations, and employees should be informed of and consent to the use of their data in such systems.

6.2 Prioritize Stress Management Interventions

Given that `stress_level` is by far the strongest predictor of burnout, organizations should invest measurably in evidence-based stress management resources. This includes access to mental health professionals, mindfulness and resilience training programs, structured workload review processes, and clear escalation pathways for employees experiencing psychological distress. These investments should be monitored for efficacy through regular pulse surveys, with data feeding back into the predictive model for ongoing recalibration.

6.3 Implement Working-Hour Monitoring and Guardrails

The strong association between `work_hours_per_week` and burnout risk justifies the introduction of systematic working-hour monitoring. Organizations should analyze team-level distributions of weekly hours and establish intervention triggers—for example, when a team member consistently exceeds 55 hours per week over a rolling four-week period. Managers should be empowered to redistribute workload or escalate resource constraints, with HR oversight to prevent structural understaffing from being absorbed silently by individual employees.

6.4 Invest in Manager Development Programs

The prominence of `manager_support` in the feature importance ranking suggests that organizations can meaningfully reduce burnout risk through targeted frontline manager development. Training programs should focus on psychologically informed management practices: recognizing early signs of employee distress, conducting supportive one-on-one conversations, facilitating workload negotiation, and creating psychological safety. Manager effectiveness in these dimensions should be incorporated into performance evaluation frameworks, with 360-degree feedback specifically addressing support behaviors.

6.5 Develop Cluster-Based Wellness Program Segmentation

Rather than applying uniform wellness initiatives across the workforce, organizations should leverage the K-Means cluster structure to design differentiated programs. For example, a cluster characterized by high stress and high work hours but strong manager support may benefit most from workload redistribution, while a cluster featuring low social support scores and high screen time may respond better to social connection and digital wellness initiatives. This segmentation approach allocates wellness resources more precisely and is more likely to achieve measurable burnout reduction outcomes.

6.6 Establish a Continuous Model Improvement Pipeline

Burnout dynamics evolve alongside organizational changes, macroeconomic conditions, and workforce demographics. The predictive model should be retrained on a rolling basis—ideally quarterly or biannually—using updated survey data. Model performance metrics (accuracy, recall on the high-burnout class) should be tracked over time to detect drift. If the high-burnout class recall declines, this may signal changes in the underlying burnout drivers that require feature re-engineering or additional data collection.

7. Discussion

7.1 Strengths of the Approach

This project demonstrates several methodological strengths. First, the breadth of the classification benchmark (covering linear, tree-based, distance-based, and probabilistic models) provides a well-rounded comparative evaluation rather than anchoring on a single algorithmic choice. Second, the deliberate removal of `burnout_score` from the feature set reflects sound data science practice: by explicitly preventing leakage, the reported accuracy figures represent genuine generalization capability rather than artifactual performance. Third, the combination of supervised and unsupervised methods enriches the analytical output beyond what classification alone would provide.

7.2 Limitations

Several limitations warrant acknowledgment. The dataset, while large, was synthetically generated and may not capture the full complexity of real organizational burnout dynamics—including confounding factors such as industry sub-sector, team size, or recent organizational change. The Label Encoding applied to ordinal and nominal categorical variables introduces a risk of spurious ordinality for non-ordered categories; One-Hot Encoding would be preferable for nominal variables in production deployments.

7.3 Improvement Pathways

Several avenues for enhancement are evident. Hyperparameter tuning (using cross-validated grid search for Random Forest (number of trees, max features, min samples per leaf) and Decision Tree (depth, splitting criterion)) would likely yield measurable accuracy gains. Gradient boosted models such as XGBoost or LightGBM, which were not implemented in this project, typically outperform Random Forest on tabular classification tasks and would represent a natural next step. The integration of time-series data (e.g., trend in stress scores over successive surveys) could substantially improve the model's ability to detect deteriorating burnout trajectories before they reach critical levels.

8. Conclusion

This project has demonstrated the viability of machine learning as a tool for early employee burnout detection in the technology sector. Across five classification algorithms trained on 23 behavioral, psychological, and organizational variables, test accuracies of 89–93% were achieved, with Logistic Regression and Random Forest performing at the top of the benchmark. The analysis conclusively identifies stress level, depression score, work hours, anxiety score, and work-life balance as the primary drivers of burnout risk; a finding with direct implications for HR strategy and organizational design.

The PCA visualization confirms that high-burnout employees occupy a distinguishable region of the feature space, supporting the feasibility of real-time risk stratification systems. The correlation analysis highlights the protective role of managerial support and work-life balance, offering clear levers for organizational intervention. Critically, the removal of the leakage variable `burnout_score` from model training ensures that reported performance figures reflect genuine predictive capability on observable inputs.

From a practical standpoint, the outputs of this project provide a foundation for HR analytics teams to move from reactive burnout management (addressing symptoms after they appear) toward predictive, proactive wellness strategies. By identifying at-risk employees earlier, organizations can deploy targeted support, reduce turnover costs, and foster a healthier, more sustainable work culture. Future work should focus on temporal modeling, improved class balancing, and the integration of real organizational data to validate and extend these findings in applied settings.

— End of Report —